FINDING b_0, b_1 WITH LEAST SQUARES [NOT EXAMINABLE]

Given b_0 and b_1 the corresponding $SSE = \sum e_i^2$

$$SSE = \sum e_i^2 = \sum (y_i - \hat{y}_i) = \sum (y_i - b_0 - b_1 x_i)^2 = f(b_0, b_1)$$

To find the minimum we set the partial derivatives equal to zero.

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = \sum \cancel{-2}\,(y_i - b_0 - b_1 x_i) = 0$$

$$\Rightarrow \underbrace{\sum y_i}_{n \cdot \bar{y}} - \underbrace{\sum b_0}_{n \cdot b_0} - b_1 \underbrace{\sum x_i}_{n \bar{x}} = 0$$

$$\Rightarrow \boxed{b_0 = \bar{y} - b_1 \bar{x}}$$

FIRST PART ✓

$$\frac{\partial f(b_0, b_1)}{\partial b_1} = \sum 2(y_i - b_0 - b_1 x_i)(-x_i)$$

$$= \sum \cancel{2}\,(y_i - \bar{y} + b_1 \bar{x} - b_1 x_i)(-x_i) = 0$$

$$= -\sum y_i x_i + \bar{y} \overbrace{\sum x_i}^{n \bar{x}} - b_1 \bar{x} \overbrace{\sum x_i}^{n \bar{x}} + b_1 \sum x_i^2 = 0$$

$$= \boxed{b_1 \big(\underbrace{\sum x_i^2 - n \bar{x}^2}_{(n-1) s_x^2} \big) = \underbrace{\sum x_i y_i - n \bar{x} \bar{y}}_{(n-1) s_{xy}}}$$

$$\Rightarrow \boxed{b_1 = \frac{s_{xy}}{s_x^2}}$$

SECOND PART ✓